

Review of Hardware Concepts for AI Infrastructure

Understanding the hardware foundation of AI performance is essential for your capstone success. We'll explore CPUs, GPUs, TPUs, NPUs, memory, interconnects, and more to give you a comprehensive review of the critical infrastructure powering modern AI systems.



CPUs: The Foundation of AI Systems

Key Characteristics

General-Purpose Compute

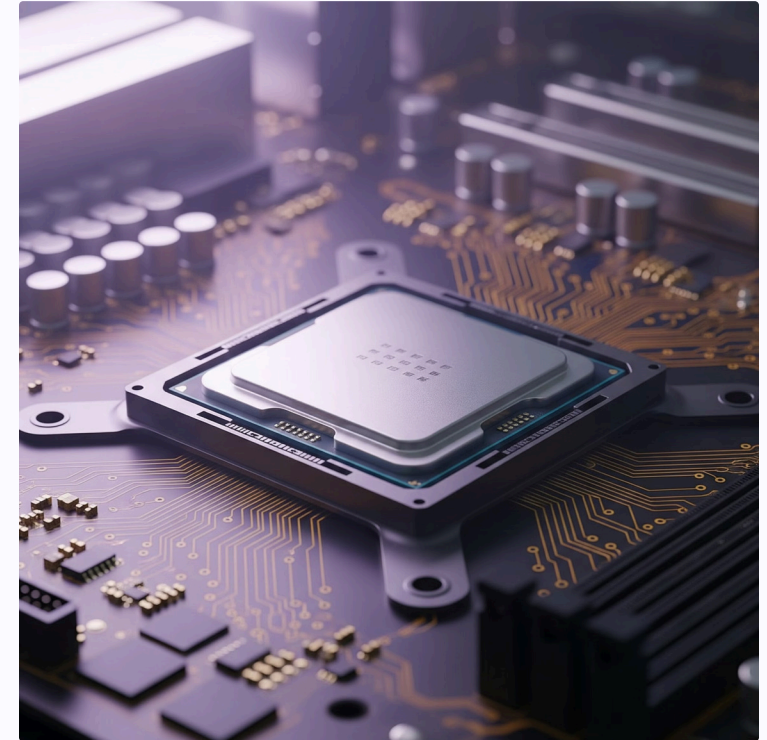
Versatile processors designed to handle various computational tasks efficiently

System Control

Excel at control logic, preprocessing, and orchestration of complex workflows

Critical Balance

Essential for maintaining system balance in AI infrastructure



Limitation: Insufficient parallel processing capabilities for large-scale AI training due to relatively few cores



GPUs: Workhorse of AI Training & Inference

Parallel Compute Architecture

Thousands of cores optimized for matrix operations, the fundamental calculations behind deep learning

Key Components

- CUDA cores for general parallel computing
- Tensor Cores for specialized AI matrix math
- FP16/FP8 precision for optimized training

Industry Leaders

NVIDIA dominates with A100 and H100 datacenter GPUs, offering unprecedented performance for large-scale AI models

TPUs: Google's AI Accelerator

Purpose-Built ASICs

Custom Application-Specific Integrated Circuits designed by Google specifically for machine learning workloads

Framework Optimization

Architected for superior performance with TensorFlow and JAX frameworks

Massive Scalability

TPU pods enable scaling to thousands of interconnected chips for training massive models

Energy Efficiency

Superior performance per watt compared to general-purpose GPUs





NPU: Enabling AI at the Edge



Mobile Integration

Apple Neural Engine and Qualcomm Hexagon processors embedded directly in consumer devices



Energy Optimization

Ultra-low power consumption enables AI inference without draining battery life



Privacy Preservation

On-device processing eliminates need to send sensitive data to cloud servers



Application Domains

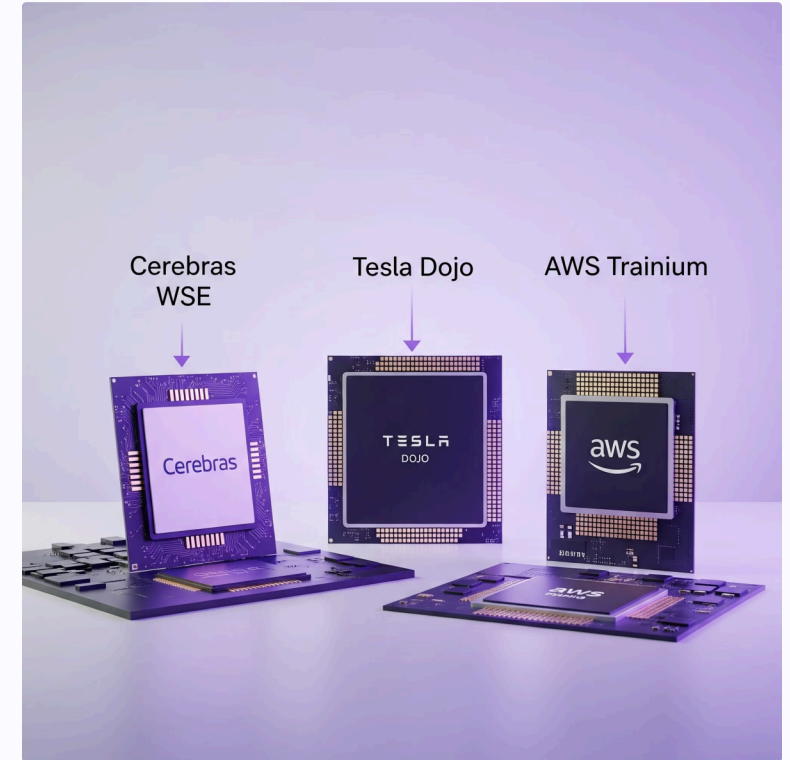
Critical for smartphones, IoT devices, robotics, and autonomous systems requiring real-time AI processing

Custom Silicon: Domain-Specific Accelerators

The industry is rapidly moving toward specialized chips optimized for specific AI workloads, offering superior performance and efficiency compared to general-purpose processors.

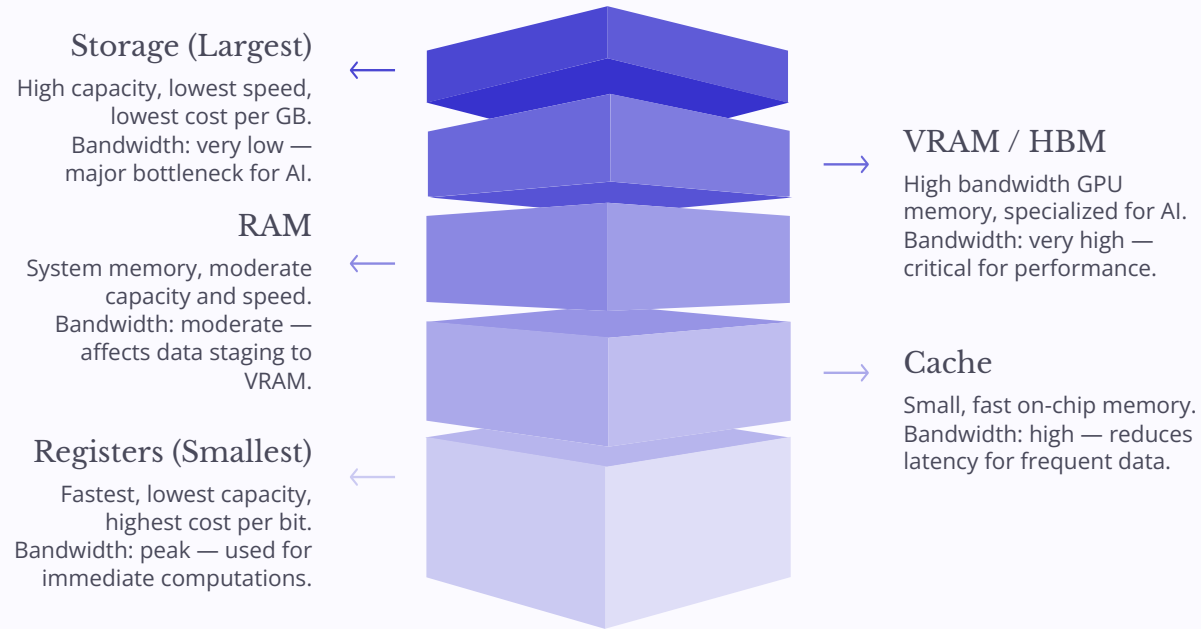
Key Innovations

- **Tesla Dojo D1:** Custom architecture optimized for computer vision training
- **AWS Trainium & Inferentia:** Purpose-built for cloud ML training and inference
- **Cerebras WSE:** Wafer-scale engine with 850,000+ cores for extreme parallelism



Trend: Domain-specific acceleration is the future of AI hardware

Memory Hierarchy: The AI Performance Pyramid



i HBM (High-Bandwidth Memory): Specialized VRAM used in GPUs and TPUs that stacks memory dies vertically to maximize bandwidth. Critical for AI model performance as it determines how quickly weights and activations can be accessed.

AI workloads are frequently memory-bound rather than compute-bound. The fundamental trade-off in memory systems is between latency, bandwidth, capacity, and cost.

Interconnects & Networking: Tying Systems Together



Chip-to-Chip

NVLink: NVIDIA's proprietary interconnect enabling 900 GB/s between GPUs

PCIe: Standard interface with lower bandwidth (64 GB/s for PCIe 5.0 x16)



Node-to-Node

InfiniBand: High-performance networking with ultra-low latency (400 Gb/s HDR)

RoCE: RDMA over Converged Ethernet for cost-effective networking



Future Directions

Optical interconnects: Silicon photonics enabling terabit-scale bandwidth

Critical for efficient distributed training across hundreds of nodes



Storage Systems: Managing AI Data

1

Local NVMe SSDs

Ultra-fast storage (7+ GB/s) for feature access and caching

Direct-attached for minimal latency during training

2

Object Storage

Amazon S3, Google Cloud Storage for massive training data lakes

Highly durable but higher access latency

3

Parallel File Systems

Lustre, BeeGFS, GPFS for HPC training clusters

Optimized for high throughput and concurrent access



The storage tier must balance three competing factors:

- Cost per terabyte
- Access speed
- Durability/reliability

Cooling & Power: The Hidden Challenge

Power Requirements

Modern AI clusters consume **megawatts** of electricity

Single H100 GPU rack: ~100 kW

Large language model training: 10+ MW

Air Cooling

Standard approach for most servers

Limitations: inefficient at high densities

Liquid Cooling

Common for high-density GPU/TPU clusters

2-3x more efficient than air cooling

Immersion Cooling

Experimental approach with servers submerged in dielectric fluid

Highest efficiency but complex implementation

- ❏ **Green AI Infrastructure:** Growing focus on sustainable computing through carbon-aware workload scheduling, renewable energy sourcing, and heat reuse systems

Performance Metrics: How to Evaluate AI Hardware

275...

Compute Performance

Measured in floating-point operations per second (FLOPS)

Example: NVIDIA H100 = 275 TFLOPS FP16

3.5...

Energy Efficiency

Performance per watt indicates how efficiently a system uses power

Critical for sustainability and operating costs

~5 ms

Inference Latency

Milliseconds per inference request

Crucial for real-time applications

2,500...

Throughput

Requests or samples processed per second

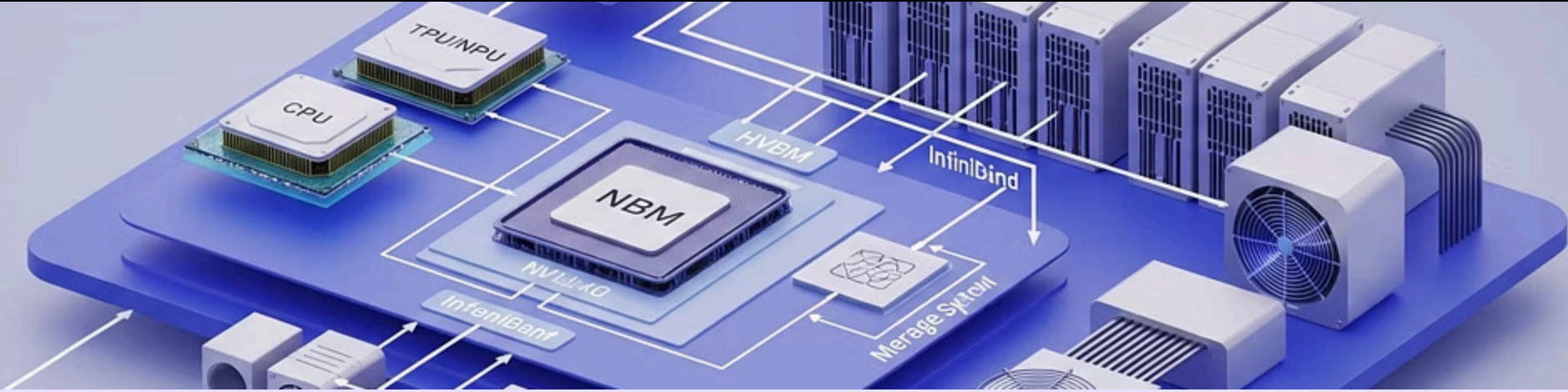
Determines overall system capacity

These metrics provide a standardized way to compare different hardware solutions and identify the best fit for specific AI workloads.

Bottlenecks in AI Hardware Systems



Understanding these bottlenecks is essential for system design. The performance of an AI system is determined by its **weakest link**, not its strongest component. Effective infrastructure requires balanced investment across all four domains.



AI Hardware Architecture: The Complete Picture

"AI hardware = tightly coupled compute + memory + interconnects."

This integrated perspective shows how all components must work together to deliver optimal AI performance. The balance between components is often more important than the raw specifications of any single element.

Key Takeaways for Your Capstone

1 GPUs Remain the AI Backbone

Despite growing competition, NVIDIA GPUs continue to dominate AI training and inference, but custom silicon solutions are rapidly gaining ground for specific workloads

2 Memory & Interconnects Often Limit Performance

High-bandwidth memory access and efficient data movement between accelerators frequently become bigger bottlenecks than raw compute capacity

3 Edge AI Powered by NPUs

Neural Processing Units are enabling AI capabilities on smartphones, IoT devices, and other edge computing platforms with stringent power constraints

4 Cooling & Sustainability Present Future Challenges

As AI models continue to grow, power consumption and cooling requirements are becoming critical engineering and environmental concerns

Discussion: Infrastructure Trade-offs

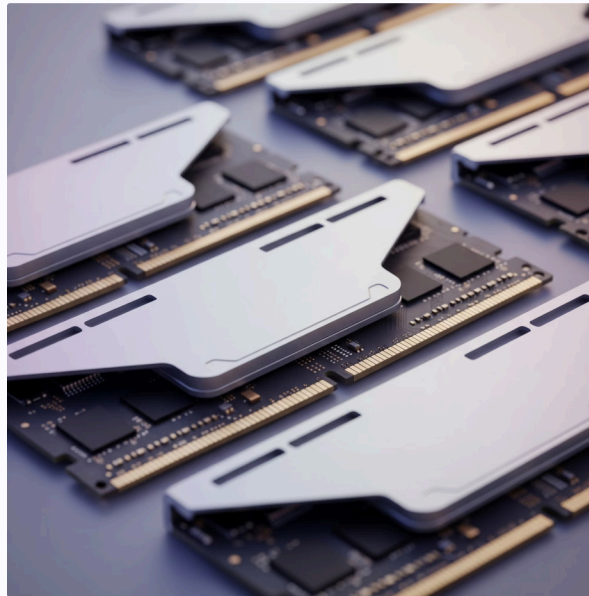
If you had to optimize AI infrastructure on a limited budget, would you invest first in GPUs (compute), HBM memory, or networking — and why?



GPU Investment

Pros: Direct compute acceleration

Cons: High upfront cost, may not help if bottlenecks are elsewhere



Memory Investment

Pros: Often the true bottleneck

Cons: Limited upgrade options



Networking Investment

Pros: Enables distributed training

Cons: Benefits only appear at scale

Consider your specific workload characteristics when making this decision. There is no one-size-fits-all answer!